

PYTHON CALCULATOR PROJECT

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A PROJECT BY NEXTHIKES IT SOLUTIONS



INTRODUCTION

- This project is a desktop calculator application created using python and Tkinter.
- The main goal was to build a calculator that is easy to use and support advanced mathematical operations.
- It works similar to a real calculator but with additional features.

OBJECTIVE

- To learn how to build a GUI application using Python.
- To perform basic and scientific calculations.
- To understand how button events work in Tkinter.
- To create a user-friendly calculator interface.

TECHNOLOGIES AND TOOLS USED

The Technologies and tools used are :

- Python programming language.
- Tkinter library for GUI design.
- Math module for scientific calculations.
- A code editor like Visual Studio Code.

FEATURES OF THIS CALCULATOR

- Basic operations like addition, subtraction, multiplication, and division.
- Scientific functions like sine, cosine, tangent, logarithm, and square root.
- A button to switch between basic to scientific modes.
- Backspace and clear button
- A history section to view previous calculations

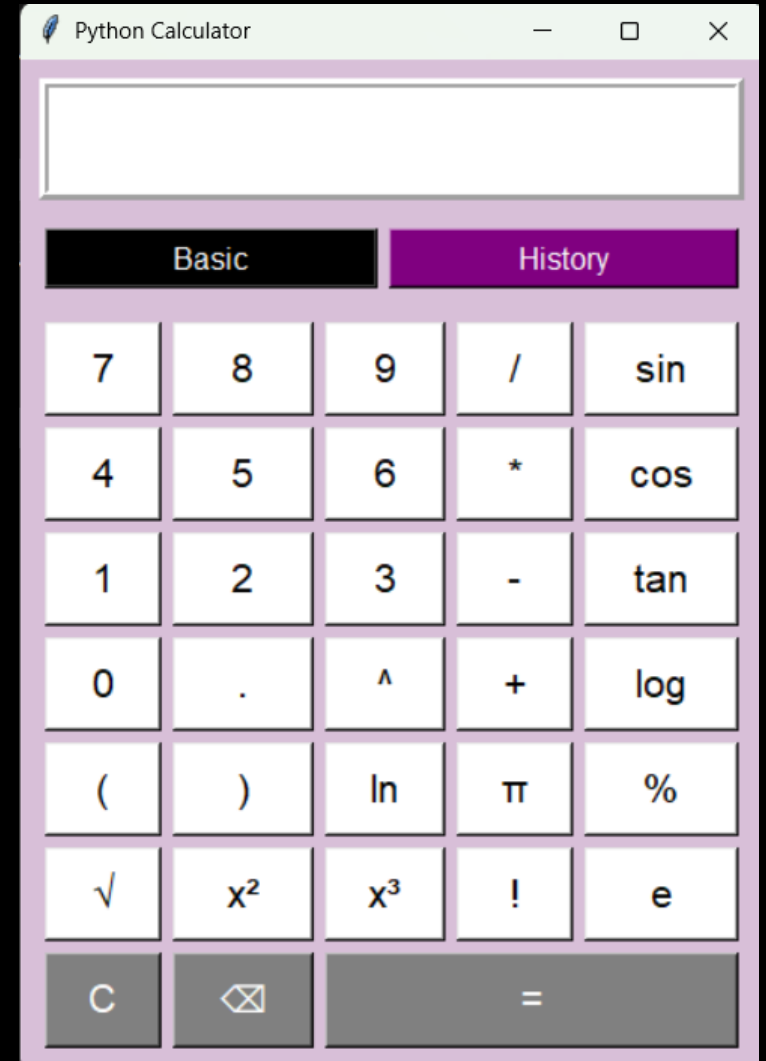
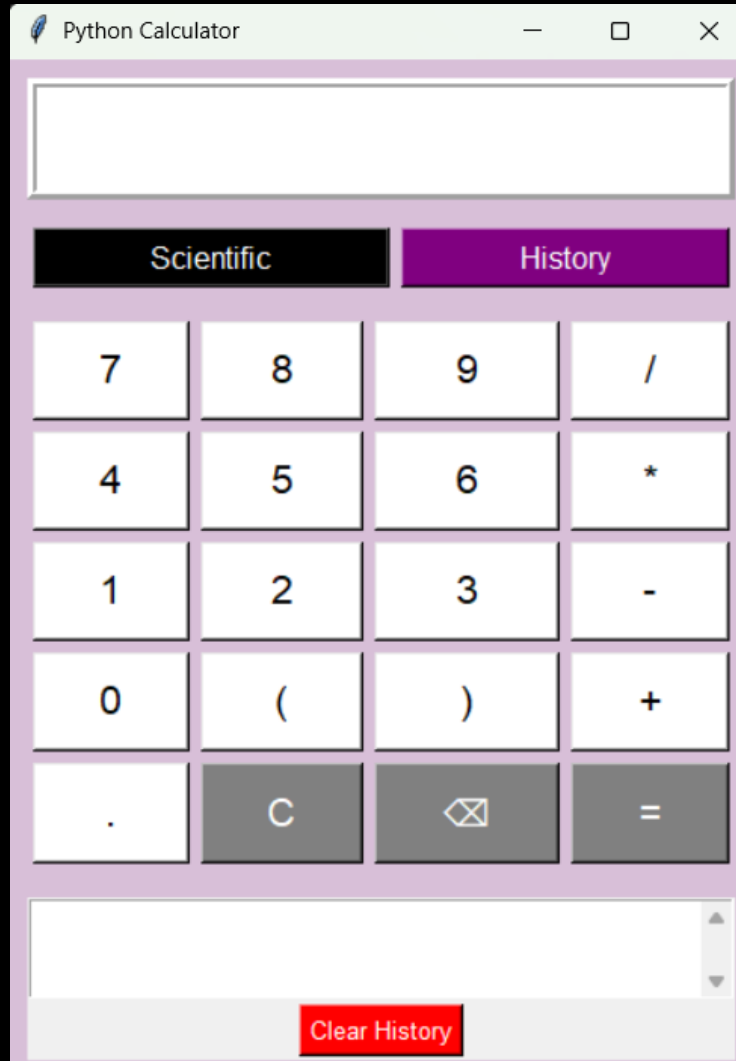
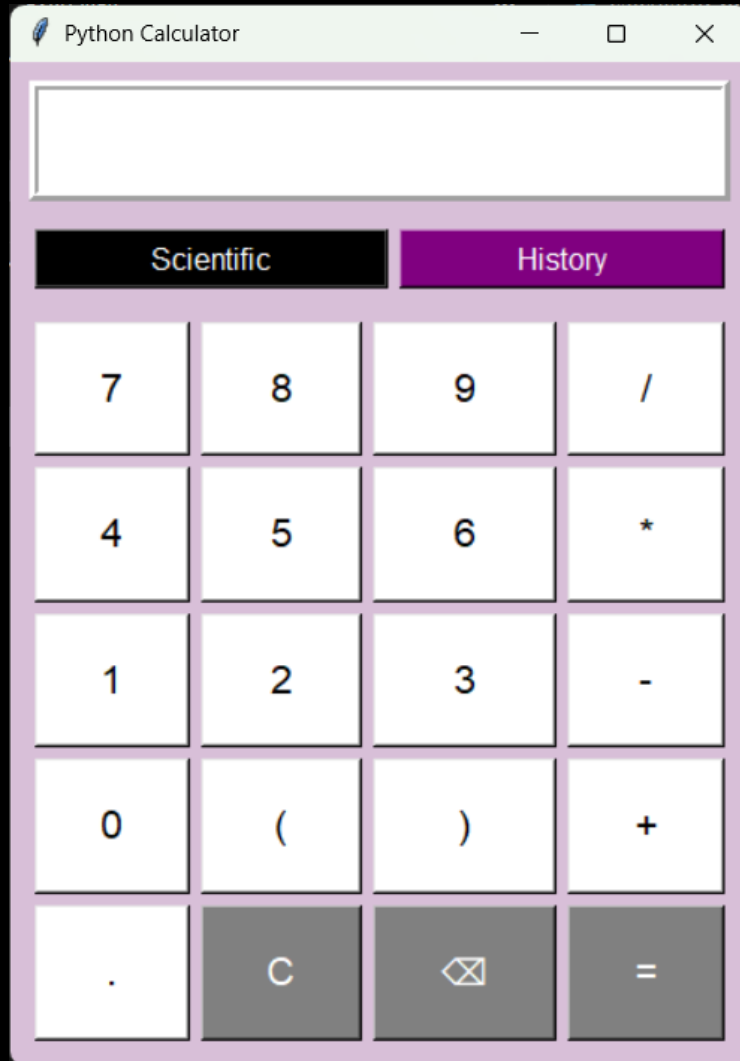
WHY THIS CALCULATOR IS DIFFERENT

- It combines basic and scientific calculators into one application
- It stores calculation history, which many simple calculators do not have
- It allows users to reuse previous calculations and clear them as well
- It is simple, interactive, and easy to understand

HOW THE CALCULATOR WORKS

- The user enters numbers and operations using buttons.
- The expression appears on the display area.
- When the equals button is pressed, the calculator processes the expression and shows the result.
- Each calculation is automatically saved in the history section for later use.

CALCULATOR PICTURE



CHALLENGES

- Adjusting the layout when switching between calculator modes
- Handling scientific symbols correctly in Python
- Preventing errors when the user enters invalid expressions
- Properly placing the buttons
- Converting mathematical symbols into Python syntax

SOLUTIONS TO THE CHALLENGES

- Using different button layouts for basic and scientific modes
- Converting scientific symbols into Python math functions
- Using error handling to prevent crashes
- Organizing the layout carefully to keep all components visible

CONCLUSION

- This project helped me strengthen my Python skills
- I learned how to build real applications using GUI
- The calculator is efficient, interactive, and user-friendly